

Bernard Lui

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Education

University of California, San Diego, San Diego, CA

Bachelor of Science in Aerospace Engineering with a specialization in Flight Dynamics and Controls

Sep. 2022 – June 2027

Grade and Honors: GPA 3.65/4.0, Provost's Honors

Relevant Coursework: Solid Mechanics, Machine Design, Controls, Fluid Dynamics, Heat Transfer, Computational Methods

Technical Skills

CAD & Design:	CATIA 3DX, SolidWorks (CAD & FEA), Onshape, Fusion 360 CAM, GD&T
Analysis & Simulation:	ANSYS Fluent (Professional & Associate Certified), MATLAB, Python (Pandas)
Manufacturing & Prototyping:	DFM/DFA, CNC and Manual Machining (mill & lathe), Additive 3D Printing, Carbon Fiber Layups (pre-preg & wet layups)
Others:	DFMEA/FMEA, Arduino, PID tuning, Integration with Sensors & Actuators, MS Office Suite

Work Experience

Tesla Inc., Energy – Safety Mechanical Design Engineer Intern, Palo Alto, CA

Jan 2026 – Sep 2026

- Designed and validated thermal propagation mitigation packages for the Megapack 2XL and 3 products, using heat transfer analysis and test data to meet thermal performance targets, comply with UL fire safety regulations, and integrate with the existing product.
- Collaborated with cross-functional teams including NPI, supply chain, and suppliers through factory line trials, DFM reviews, and DFMEA to drive products from concept toward production.
- Led the design and fabrication of the first battery module coupon, kickstarting a multi-year program to enable rapid prototyping and accelerated safety and abuse testing, and developed new assembly fixtures compatible with existing lab capabilities.
- Performed first-principles hand calculations, considering foam compression dynamics, thermal representativeness, bolt mechanics and electrical isolation, verified the coupon structure using ANSYS Mechanical FEA.
- Conducted FMEA studies for various tests; collaborated closely with the test engineering team to prioritize high-risk items based on occurrence and severity and create test build maps. Used FEA-identified weak points to define the module drop test plan, execute tests, and analyze compliance and abuse testing results to drive design feedback.

UC San Diego Mechanical and Aerospace Department, Instructional Assistant, San Diego, CA

Sep. 2023 – Present

- Refine a newly designed project for an introductory aerospace class, completing prototype builds and testing of the control-line aircraft, addressing any problems identified or potential improvements along the way.
- Mentor 75+ students in lab groups through the full project cycle, troubleshooting issues on the spot, guiding them through worksheet calculations questions.
- Assist professor by grading homework and exams, providing students with timely feedback in thermodynamics and fluid mechanics topics, and provide insights during weekly meetings on patterns in student performance.

Project Experience

Triton Racing Formula SAE, Aerodynamics Lead Engineer, San Diego, CA

Oct 2024 – Jan 2026

- As the aerodynamics and composite lead engineer:
 - Led design and analysis of TR-26 aerodynamic package; optimized and validated airfoil selections through CFD, improving aerodynamic efficiency by 10%.
 - Managed full aerodynamic design cycle for an 11-member sub-team, delegated tasks, oversaw procurement from various vendors, and in-house manufacturing of 20+ components to ensure on-schedule delivery of the aero package.
 - Developed pre-preg wing spars through a MATLAB beam bending analysis, calculated stiffness and structural performance under thermal and mechanical loads; conducted DFM reviews with lab technicians, and validated prototypes through iterative testing.
- As an aerodynamics and suspension member:
 - Designed and optimized TR-24 suspension rocker assembly in SolidWorks with FEA and Fusion360 generative design, achieving 20% weight reduction while maintaining manufacturability; prototyped and manufactured via CNC machining.
 - Optimized rear wing design with CFD with high-lift, good post-stall performance airfoils and integrated with DeDion suspension assembly for the team's first EV; manufactured parts via carbon fiber wet and pre-preg layups, and mill and lathe machining.

Personal Project

Computational Structural Analysis of an Aircraft Wing Under Aerodynamic Loads

Jan. 2025 – Mar. 2025

- Modeled aerodynamic forces acting on a Cessna 172 wing using exact and approximate equations, analyzing material behavior including spar stresses, deflection, and torsional forces under various load conditions.
- Applied numerical methods in MATLAB to evaluate wing behavior under worst-case loads and determined key design parameters such as wing length, skin thickness, and spar thickness.